

Chapter 5 Discriminant Analysis

5. 4. From the previous experience, the humidity difference(湿度差) between today and yesterday, the difference between air pressure and temperature (压温差) today are two important factors for predicting whether it rains or not tomorrow. A batch of sample data is now collected and listed in the table below.

π_1 (rain)		π_2 (not rain)	
x_1 (湿度差)	x_2 (压温差)	x_1 (湿度差)	x_2 (压温差)
-1.9	3.2	0.2	6.2
-6.9	10.4	-0.1	7.5
5.2	2.0	0.4	14.6
5.0	2.5	2.7	8.3
7.3	0.0	2.1	0.8
6.8	12.7	-4.6	4.3
0.9	-15.4	-1.7	10.9
-12.5	-2.5	-2.6	13.1
1.5	1.3	2.6	12.8

3.8	6.8	-2.8	10.0
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Now we observed $x_1 = 0.6$, $x_2 = 0.3$. Assume that there is the same covariance matrix for the two groups π_1 and π_2 . Please answer the following questions:

- 1) To give the distance discrimination rule and forecast whether it will rain or not tomorrow.
- 2) Assume $\mathbf{x} = (x_1, x_2)'$ follows a bivariate normal distribution in the above two groups and the prior probability $p_1 = 0.4$, $p_2 = 0.6$, try to predict whether it will rain or not tomorrow based on the maximum posterior probability criterion (最大后验概率准则).

Hints: The maximum posterior probability criterion $P(\pi_i|\mathbf{x}) =$

$$\frac{\exp(\mathbf{I}_i' \mathbf{x} + c_i + \ln p_i)}{\sum_{i=1}^k \exp(\mathbf{I}_i' \mathbf{x} + c_i + \ln p_i)}, i = 1, 2 \text{ 其中 } \mathbf{I}_i = S_p^{-1} \bar{\mathbf{x}}_i, c_i = -\frac{1}{2} \bar{\mathbf{x}}_i' S_p^{-1} \bar{\mathbf{x}}_i, i = 1, 2.$$

- 3) If you are thinking about arranging an activity for tomorrow, but this activity is not suitable for rainy days. Assume that $c(2|1) = 3c(1|2)$, so should you schedule this activity for tomorrow today?

It is known that

$$n_1 = n_2 = 10, n = 20, \bar{\mathbf{x}}_1 = (0.92, 2.10)', \bar{\mathbf{x}}_2 = (-0.38, 8.85)',$$

$$\bar{\mathbf{x}} = \frac{1}{2}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) = (0.27, 5.475),$$

(1')

令 $\mathbf{x}_0 = (0.6, 0.3)$,

$$S_1 = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)(x_{1j} - \bar{x}_1)' = \begin{pmatrix} 40.8973 & 6.3644 \\ 6.3644 & 59.6867 \end{pmatrix},$$

$$S_2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)(x_{2j} - \bar{x}_2)' = \begin{pmatrix} 6.2084 & -0.3378 \\ -0.3378 & 18.5450 \end{pmatrix},$$

$$S_p = \frac{1}{2}(S_1 + S_2) = \begin{pmatrix} 22.55285 & 3.0133 \\ 3.0133 & 39.11585 \end{pmatrix}, S_p^{-1} =$$

$$\begin{pmatrix} 0.04289 & -0.0033 \\ -0.0033 & 0.0258 \end{pmatrix},$$

5.7 The following table shows the data of breakfast porridge produced by three American manufacturers I, II, III. The breakfast porridge (粥) brands were grouped according to the manufacturers. The indicators of each brand were: calories (x_1), protein (x_2), fat (x_3), sodium (x_4), fiber (x_5), carbohydrate (x_6), sugar (x_7) and potassium (x_8). We collected the sample from the three groups as follows:

编号	组别	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
1	I	110	2	2	180	1.5	10.5	10	70
2	I	110	6	2	290	2.0	17.0	1	105
3	I	110	1	1	180	0.0	12.0	13	55
4	I	110	1	1	180	0.0	12.0	13	65
5	I	110	1	1	280	0.0	15.0	9	45
6	I	110	3	1	250	1.5	11.5	10	90
7	I	110	2	1	260	0.0	21.0	3	40
8	I	110	2	1	180	0.0	12.0	12	55
9	I	100	2	1	220	2.0	15.0	6	90
10	I	130	3	2	170	1.5	13.5	10	120
11	I	100	3	2	140	2.5	10.5	8	140
12	I	110	2	1	200	0.0	21.0	3	35
13	I	140	3	1	190	4.0	15.0	14	230
14	I	100	3	1	200	3.0	16.0	3	110
15	I	110	1	1	140	0.0	13.0	12	25
16	I	100	3	1	200	3.0	17.0	3	110
17	I	110	2	1	200	1.0	16.0	8	60
18	II	70	4	1	260	9.0	7.0	5	320
19	II	110	2	0	125	1.0	11.0	14	30
20	II	100	2	0	290	1.0	21.0	2	35
21	II	110	1	0	90	1.0	13.0	12	20
22	II	110	3	3	140	4.0	10.0	7	160
23	II	110	2	0	220	1.0	21.0	3	30
24	II	110	2	1	125	1.0	11.0	13	30
25	II	110	1	0	200	1.0	14.0	11	25
26	II	100	3	0	0	3.0	14.0	7	100
27	II	120	3	0	240	5.0	14.0	12	190
28	II	110	2	1	170	1.0	17.0	6	60
29	II	160	3	2	150	3.0	17.0	13	160
30	II	120	2	1	190	0.0	15.0	9	40
31	II	140	3	2	220	3.0	21.0	7	130

续表

编号	组别	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
32	II	90	3	0	170	3.0	18.0	2	90
33	II	100	3	0	320	1.0	20.0	3	45
34	II	120	3	1	210	5.0	14.0	12	240
35	II	110	2	0	290	0.0	22.0	3	35
36	II	110	2	1	70	1.0	9.0	15	40
37	II	110	6	0	230	1.0	16.0	3	55
38	III	120	1	2	220	0.0	12.0	12	35
39	III	120	1	2	220	1.0	12.0	11	45
40	III	100	4	2	150	2.0	12.0	6	95
41	III	50	1	0	0	0.0	13.0	0	15
42	III	50	2	0	0	1.0	10.0	0	50
43	III	100	5	2	0	2.7	1.0	1	110

Assume that there is a common covariance matrix for the tree population groups.

Questions:

- 1) Try to give Fisher discriminant, and report discrimination results of the samples for different groups I, II, III in the above table with the following table

True group \ Discriminated as	I	II	III
I			
II			
III			

- 2) Report the misclassification probabilities based on the above table (Reuse of samples is allowed).

3) Draw the two discriminant scores of all brands into a scatter diagram, and use different symbols to represent different manufacturers.

Hints: Use centralized Fisher discriminant,

$$y_j = \mathbf{t}_j'(\mathbf{x} - \bar{\mathbf{x}}), j = 1, 2, 3$$

$$\bar{y}_{ij} = \mathbf{t}_j'(\bar{\mathbf{x}}_i - \bar{\mathbf{x}}), \quad j = 1, 2, \dots, n_i$$

Where $\bar{\mathbf{x}} = (107.9070, 2.4651, 0.9767, 180.4651, 1.7140, 14.2558, 7.6047, 84.4186)$

and the first two eigenvectors of $E^{-1}B$ need in the Fisher discriminant are

$$\begin{aligned} \mathbf{t}_1 &= (0.0223, 0.3691, -0.8377, -0.0008, 1.4203, 0.2022, 0.1952, -0.0307)', \\ \mathbf{t}_2 &= (-0.0454, 0.3324, 0.3865, 0.0060, -1.0400, 0.2039, 0.2353, 0.0270)' \end{aligned}$$

Fisher discriminant is given as follows:

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{j=1}^2 (y_j - \bar{y}_{lj})^2 = \min_{1 \leq i \leq 3} \sum_{j=1}^2 (y_j - \bar{y}_{ij})^2$$

5.8 使用 SAS 软件对习题 5.5 中的数据做逐步判别 (取 $\alpha=0.05$) (选做题), 数据及 SAS 程序已放在 BB 上, 选做改题的目的是帮助自己熟悉上 SAS 程序及分析。

Deadline: 14: 00, April 14 (Monday, Week 9) , 2025.